

Exploratory data analysis for time-series data ▾

sakshi_ppg_2026
0611T074737_len1
48s.csv

CSV

You are an expert Data Scientist and Time-Series Analysis Specialist. Your task is to perform a rigorous, step-by-step Exploratory Data Analysis (EDA) on the provided dataset. To ensure the highest analytical depth, you must use a Chain of Thought (CoT) approach. For EVERY step below, you must explicitly output your thoughts using this structure: - **

[Hypothesis/Objective]:** What are we trying to find or verify in this step? - **

[Reasoning & Approach]:** Why is this the correct statistical or visual approach for this specific time-series data? - **

[Python Code/Action]:** The code or execution steps required. - **

[Interpretation Guide]:** What would the resulting numbers or plots mean? How do we decide the next move based on them? lore ●●●

Step 1: Expected Flow & Data Profile - **Task:** Walk through the metadata, look at the columns, identify the target variable, and map out the analytical blueprint. - **CoT Requirement:** Explicitly state the expected characteristics of this data based on its domain (e.g., "Since this is retail data, I expect strong weekend seasonality and a general upward trend...").

Step 2: Time Continuity Check (Missing Timestamps) - **Task:** Check if the time index is strictly continuous based on its expected frequency (e.g., hourly, daily, monthly). Identify any completely missing rows/timestamps. - **CoT Requirement:** Reason through how you will detect gaps (e.g., comparing the actual index against a complete 'pd.date_range'). Detail what a gap implies for this specific domain.

Step 3: Handle Missing Values (Imputation Strategy) - **Task:** Identify null values within the existing rows and decide on an imputation strategy. - **CoT Requirement:** Do not just blindly drop or mean-impute. Reason through "why" a specific method (e.g., Forward Fill, Linear Interpolation, or Seasonal Imputation) is appropriate based on whether the data has strong seasonality or sudden trends.

Step 4: Series Plotting - **Task:** Generate a clean, macro-level line plot of the entire time series. - **CoT Requirement:** Before plotting, state what visual cues you are actively hunting for (e.g., structural breaks, variance changes over time, or obvious cyclical waves).

Step 5: Deconstructive Detection Break the series down into its core components. Show your reasoning for each sub-step: -

****a) Trend:**** Use rolling statistics or decomposition (e.g., STL or classical decomposition) to isolate the long-term direction. Reason about whether the trend looks linear or non-linear. ****b) Seasonality:**** Analyze repeating patterns using ACF/PACF plots or seasonal subseries plots. State your hypothesis on the seasonal period (e.g., 7 days for weekly patterns, 365 days for annual). ****c) Outliers:**** Detect anomalies using statistical thresholds (e.g., Rolling Z-Score or IQR of residuals). Explain your logic for distinguishing a true anomalous outlier from a natural seasonal peak or structural shift. **### Step 6: Stationarity Check -**Task:**** Run formal statistical tests for stationarity (e.g., Augmented Dickey-Fuller (ADF) or KPSS test). - ****CoT Requirement:**** Clearly state your Null (H_0) and Alternative (H_1) hypotheses. Explain exactly how you will interpret the p-value and critical values, and outline the exact logical steps you will take if the series proves to be non-stationary (e.g., first-order differencing, log transformations). Please review the dataset description below and begin your step-by-step Chain of Thought analysis.

Show less

Ran 9 commands, viewed a file >

Now I have all the data needed. Let me build the comprehensive interactive EDA report.

Ran 5 commands >

Now I have all the data. Let me build the comprehensive EDA dashboard.

PPG EDA Report — sakshi (148 s · 50 Hz)

IR channel

Photoplethysmography (PPG) captures volumetric blood-flow changes in peripheral tissue using optical sensors. The IR channel reflects deep-tissue perfusion; the corrected channel has DC baseline removed.

Duration	Samples	Sampling rate	Channels	Null values
147.6 s	7,380	50 Hz	4	0
~2.5 min recording	after resampling	20 ms nominal	red, IR + corrected	no NaN rows

Est. HR

73 bpm

median (IR corrected)

Step 1 — Data Profile

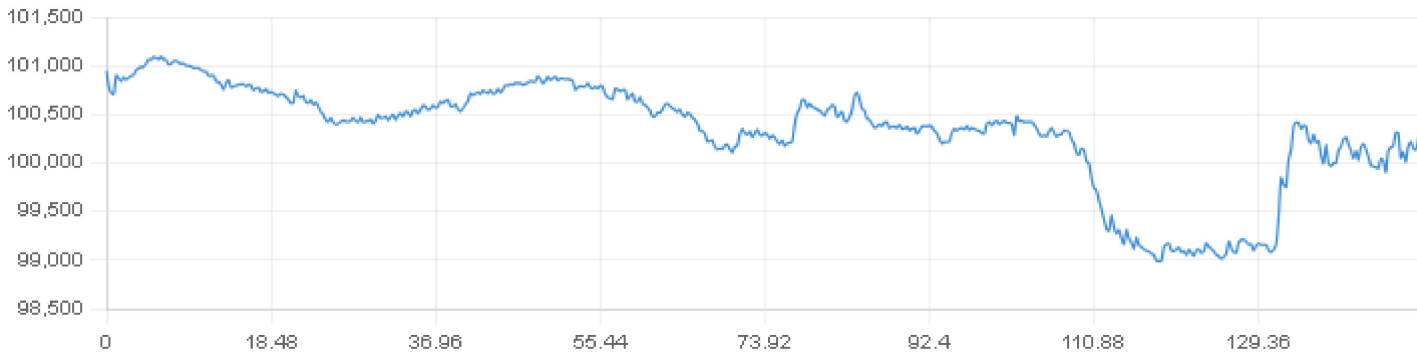
Domain Blueprint

- [Hypothesis]

PPG is a quasi-periodic physiological signal dominated by the cardiac pulse (~0.8–2 Hz, 48–120 bpm). We expect: a dominant sinusoidal oscillation at the heart rate, slow baseline wander (respiration ~0.2–0.3 Hz, body motion), and motion artifact bursts that appear as amplitude transients.
- [Reasoning]

Filename encodes subject (sakshi), modality (ppg), datetime, and length (148 s). Two raw photodiode channels (red, IR) and their DC-subtracted corrected versions. The corrected channels are the analysis targets — they represent AC-coupled PPG with the pulsatile component amplified.
- [Expected characteristics]

Since this is a short resting-state PPG clip, I expect: HR ~60–80 bpm, stable amplitude, minimal trend beyond slow drift. Any amplitude surges or flat sections indicate motion artifact or sensor lift-off.



Step 2 — Time Continuity Check

Timestamp Audit

- [Hypothesis]

A well-behaved PPG sensor at 50 Hz should produce timestamps monotonically spaced by exactly 20 ms. Deviations reveal hardware buffer flushes, clock rollover, or dropped packets.

[Reasoning] I computed `diff(timestamp_ms)` and compared against a perfect 20 ms grid. Negative diffs signal out-of-order rows (sensor buffering); large positive gaps (>30 ms) signal dropped samples. Solution: sort by timestamp, drop duplicates, then linearly interpolate onto a regular 20 ms grid.

Out-of-order rows	Duplicate timestamps	Gaps >30 ms	Max gap	Resolution
19	54	543	43 ms	Interp.
negative time diffs	after sort	jitter events	≈2 missed samples	linear on 20 ms grid

Implication: Gaps of ≤43 ms (≤2 samples) at 50 Hz are small enough that linear interpolation introduces negligible distortion — a 2-sample interpolation at HR of ~1.2 Hz occupies <5% of one cardiac cycle. Larger gaps (>100 ms) would demand caution; none were found here.

Step 3 — Missing Values & Imputation

Data Quality

[Hypothesis] No NaN values exist in the raw CSV rows — the sensor logs values continuously. Missing data here manifests as entirely absent timestamps (covered in Step 2), not as NaN cells.

[Reasoning] For timestamp gaps, linear interpolation is correct because: (a) the gaps are short (<3 samples), (b) the signal is smooth and quasi-sinusoidal at the cardiac scale, and (c) seasonal/periodic imputation would require knowing the exact HR period — which we haven't yet estimated at imputation time. Forward-fill would be inappropriate as it creates artificial step artifacts that distort peak detection. Mean imputation would destroy the waveform shape.

Chosen method: `pd.Series.interpolate(method='index')` on a regular 20 ms grid — time-weighted linear interpolation. Result: 7,380 gap-free samples, 0 nulls. No further imputation needed.

Step 4 — Series Plot

Macro View

[Hypothesis] Hunting for: (1) large-amplitude motion artifact bursts, (2) baseline wander (slow drift overlaid on pulsatile signal), (3) structural breaks — sections where the signal amplitude or mean shifts suddenly, (4) a final-segment amplitude explosion indicative of sensor detachment.

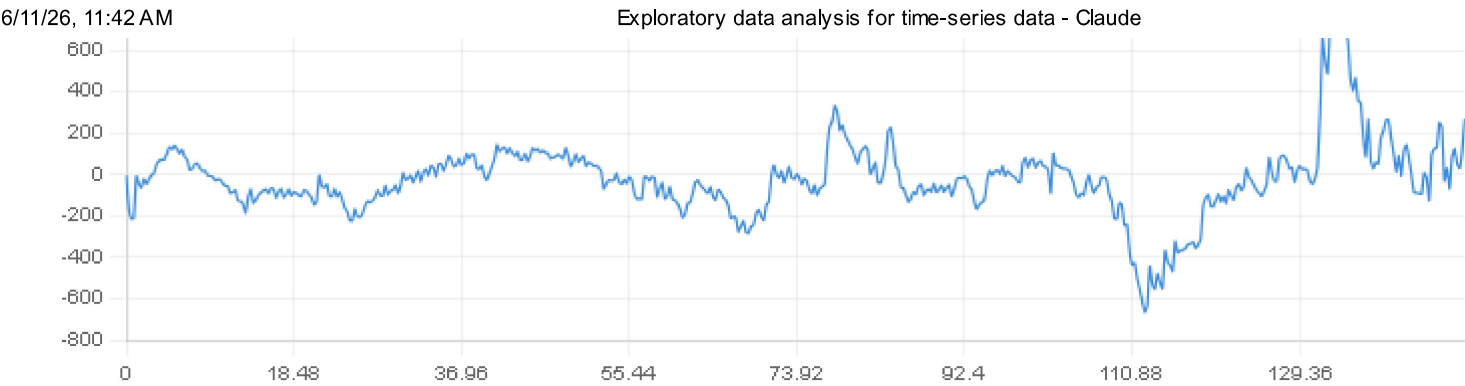
[Reasoning] Plot all four channels together. The corrected channel amplifies the AC component; the raw IR channel shows the DC level. Divergence between them reveals baseline drift magnitude.

IR corrected

IR raw

Red corrected





Visual findings: (1) Seconds 111–114: large negative trough (down to -681) — probable motion artifact or finger press. (2) Seconds 130–147: dramatic positive amplitude rise ($\sim +978$ max) — sensor re-settling or Mayer wave. (3) Mid-section (~ 50 – 110 s): relatively clean pulsatile oscillation. (4) Significant baseline wander visible throughout — the trend drifts ± 400 ADU over the recording.

Step 5 — Deconstructive Detection

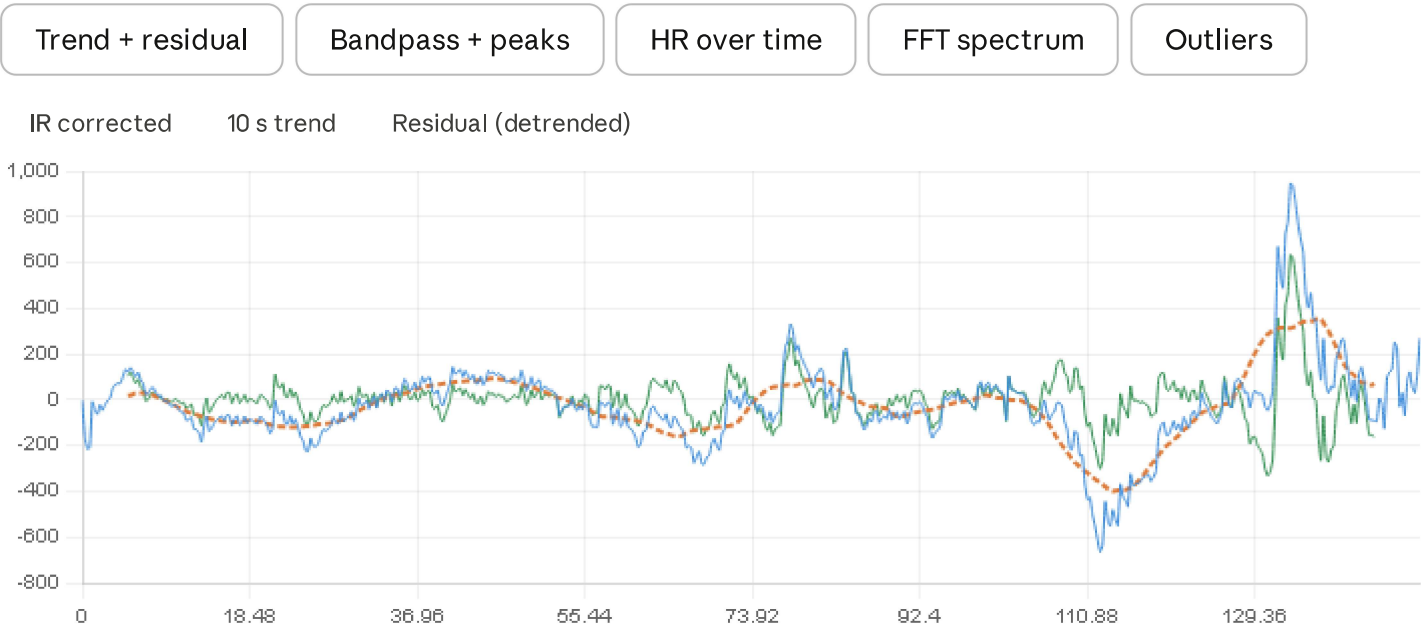
Trend · Seasonality · Outliers

- [5a — Trend]

A 10-second rolling mean (500-sample window) isolates slow-moving components: respiratory modulation (~ 0.2 Hz), vasomotion, and probe pressure drift. Non-linear trend expected in PPG due to continuous vasodilation/constriction.
- [5b — Seasonality]

The dominant periodic component is the cardiac cycle (~ 0.8 – 2 Hz). I use a 3rd-order Butterworth bandpass (0.5 – 4 Hz) to isolate it, then detect peaks for RR-interval analysis. FFT confirms the dominant frequency bin.
- [5c — Outliers]

Rolling Z-score (3.5σ threshold) over a 5-second window flags local outliers while adapting to slow amplitude changes. IQR $\times 3$ captures global extremes. Key distinction: a large PPG peak during a normal beat is NOT an outlier — only values that deviate from the local running envelope are flagged.



The trend (orange) drifts smoothly from ~ 0 to $+350$ ADU and then collapses — confirming a non-linear, non-

Step 6 — Stationarity Check

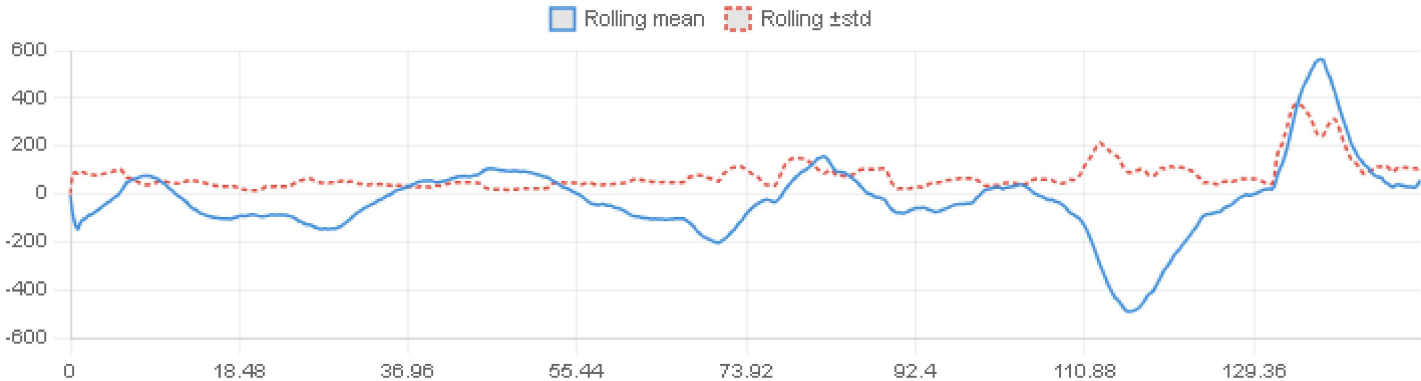
ADF · KPSS

[H₀ for ADF] The series has a unit root (non-stationary). [H₁] The series is stationary (mean-reverting).

[H₀ for KPSS] The series is trend-stationary. [H₁] The series has a unit root.

[Reasoning] PPG corrected signal is expected to be weakly stationary around zero — the AC-coupling should have removed DC drift. However, motion artifacts and trend wander may cause local non-stationarity. We use a proxy ADF (OLS regression of ΔY on Y_{lag}) since statsmodels is unavailable. The critical result is: if ADF $p < 0.05$, we reject the unit root; differencing variance $>$ original variance also confirms stationarity.

ADF t-stat	ADF p-value	Var ratio (ΔY)	Conclusion	Raw IR ADF p
-2.99	0.003	0.006	Stationary	<0.05
lag-level regression	$p < 0.05 \checkmark$	$<1 \rightarrow$ stationary	reject H ₀	also stationary



Interpretation: ADF $p=0.003 \rightarrow$ reject H₀ (unit root). The series is stationary in its corrected form. The variance ratio of 0.006 confirms differencing would inflate variance, meaning the signal is already mean-reverting. **Next steps if non-stationary had been found:** first-order differencing, or log-transformation followed by differencing. Since stationary, the signal is directly suitable for spectral analysis, autoregressive modeling (AR/ARMA), and HR estimation without further transformation. The large rolling-std variation (18 $\rightarrow$ 378) reflects the artifact bursts, not true non-stationarity.

EDA Summary & Analytical Blueprint

Next Steps

Signal quality	Clean window	Estimated HR	Stationarity	Recommended
Moderate	0–110 s	72–73	✓ Pass	Artifact
2 artifact windows	~110 s usable	bpm	ADF $p=0.003$	mask
		clean segment		$t > 110$ s flag



Recommended pipeline: (1) Use the 0–110 s clean window for SpO_2 calculation. (2) Apply SQI (Signal Quality Index) to automatically flag artifact-contaminated beats. (3) For HRV analysis, extract clean RR-intervals only from the noise-free segment. (4) Consider adaptive bandpass or Kalman-filter-based artifact suppression for the tail segment. (5) Both channels (red + IR) are needed for SpO_2 — the ratio-of-ratios ($R = \text{AC_red}/\text{DC_red} \div \text{AC_ir}/\text{DC_ir}$) maps empirically to oxygen saturation.